

⁹Be (Be I), nuclear spin $I = 3/2$. Energy levels and transitions below 2 μm : 15 levels, 24 lines.

Be⁺ ionisation limit 75192.64 cm⁻¹ = 9.32270 eV (132.992 nm)

Arrow colour = vacuum wavelength (nm)



300 400 500 600 700 800 1000 1200 1500 2000



Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J er J' \rangle $ (eao)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$2s^2\ ^1S_0 - 2s2p\ ^1P_1^o$	234.9323	234.8604	1276.0801	3.255	5.520e+08	-	-	
$2s2p\ ^3P_0^o - 2s3d\ ^3D_1$	249.5295	249.4542	1201.4311	1.569	1.070e+08	-	-	
$2s2p\ ^3P_1^o - 2s3d\ ^3D_2$	249.5333	249.4580	1201.4128	2.350	1.440e+08	-	-	
$2s2p\ ^3P_2^o - 2s3d\ ^3D_1$	249.5333	249.4580	1201.4127	1.360	8.040e+07	-	-	
$2s2p\ ^3P_2^o - 2s3d\ ^3D_2$	249.5478	249.4726	1201.3428	2.714	1.920e+08	-	-	
$2s2p\ ^3P_1^o - 2p^2\ ^3P_2$	265.1250	265.0461	1130.7588	2.208	1.060e+08	-	-	
$2s2p\ ^3P_0^o - 2p^2\ ^3P_1$	265.1347	265.0558	1130.7176	1.973	1.410e+08	-	-	
$2s2p\ ^3P_1^o - 2p^2\ ^3P_1$	265.1390	265.0601	1130.6992	1.710	1.060e+08	-	-	
$2s2p\ ^3P_2^o - 2p^2\ ^3P_2$	265.1414	265.0625	1130.6888	3.819	3.170e+08	-	-	
$2s2p\ ^3P_1^o - 2p^2\ ^3P_0$	265.1488	265.0699	1130.6575	1.973	4.230e+08	-	-	
$2s2p\ ^3P_2^o - 2p^2\ ^3P_1$	265.1554	265.0765	1130.6292	2.204	1.760e+08	-	-	
$2s2p\ ^3P_0^o - 2s3s\ ^3S_1$	332.1969	332.1013	902.4542	0.961	1.500e+07	-	-	
$2s2p\ ^3P_1^o - 2s3s\ ^3S_1$	332.2036	332.1081	902.4358	1.664	5.710e+07	-	-	
$2s2p\ ^3P_2^o - 2s3s\ ^3S_1$	332.2294	332.1338	902.3658	2.150	8.510e+07	-	-	
$2s2p\ ^1P_1^o - 2s3s\ ^1S_0$	825.6344	825.4075	363.1056	3.064	3.380e+07	-	-	
$2s3s\ ^3S_1 - 2s3p\ ^3P_2^o$	1464.7945	1464.3943	204.6652	9.526	1.170e+07	-	-	

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST. $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.